Connected High-Event Registry

Analytic ledger replacement, Packet C

Abstract

This note replaces the exact-executable arithmetic used in the high spherical-shell staircase by displayed rational witnesses. It contains the complete seventeen-face localization registry, the three algebraic splits, the three dashed (impossible) cap cells, the five global checkpoint payments, the seven conditional payments, and the corrected cap–74 path. All comparisons are strict rational comparisons after the two elementary bounds $333/106 < \pi < 22/7$. In particular, no decimal approximation, interval package, or external ledger is a logical input to this packet.

1 Definitions and strict proxy convention

For $0 < \rho < 1$, put

$$z = z_\rho := \frac{\pi}{1-\rho}, \qquad x = x_\rho := z_\rho^2, \qquad k_m(\rho) := \sqrt{x_\rho + m(m+1)}. \tag{1}$$

The three-dimensional Weyl term is

$$\mathcal{W}(\rho, K) := \frac{2}{9\pi}(1-\rho^3)K^3. \tag{2}$$

At $K = k_m(\rho)$, the product part of the strict channel proxy for the mode (ℓ, n), $n \geq 2$, is

$$n^2 x + \ell(\ell+1) < x + m(m+1).$$

It is therefore governed by

$$\Delta_{m;n,\ell}(\rho) := m(m+1) - \ell(\ell+1) - (n^2-1)x_\rho. \tag{3}$$

The channel is product-possible only when $\Delta_{m;n,\ell} > 0$. Equality is excluded because the spectral proxy is strict. Since

$$x'_\rho = \frac{2\pi^2}{(1-\rho)^3} > 0, \qquad \Delta'_{m;n,\ell} = -(n^2-1)x'_\rho < 0, \tag{4}$$

a positive witness at r controls the side $\rho \leq r$, while a negative witness controls the side $\rho \geq r$.

We write H for the largest first-mode angular index retained in a cap cell and h for the largest second-mode angular index. A complete block through H has multiplicity $(H+1)^2$, because $\sum_{\ell=0}^{H}(2\ell+1) = (H+1)^2$.

2 The only transcendental enclosure

Lemma 1 (Rational enclosure of π)**.** *One has*

$$\pi_- := \frac{333}{106} < \pi < \frac{22}{7} =: \pi_+. \tag{5}$$

Proof. For $0 < u < 1$, the alternating expansion of $\arctan u$ lies between each pair of consecutive partial sums. The tangent addition formula gives

$$\tan\left(4\arctan\frac{1}{5} - \arctan\frac{1}{239}\right) = 1;$$

indeed, $\tan(4\arctan(1/5)) = 120/119$, so both the numerator and denominator in the last tangent quotient are $28561/(119 \cdot 239)$. The angle lies in $(0, \pi/2)$, and hence Machin's identity follows:

$$\pi = 16\arctan\frac{1}{5} - 4\arctan\frac{1}{239}.$$

Using an even partial sum for the positive term and an upper partial sum for the subtracted term gives

$$\begin{aligned} \pi &> 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5} - \frac{1}{7\cdot 5^7}\right) - \frac{4}{239}, \\ \text{[right side]} - \frac{333}{106} &= \frac{3418213}{41563593750} > 0. \end{aligned}$$

Likewise,

$$\begin{aligned} \pi &< 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3} + \frac{1}{5\cdot 5^5}\right) - 4\left(\frac{1}{239} - \frac{1}{3\cdot 239^3}\right), \\ \frac{22}{7} - \text{[right side]} &= \frac{1845738322}{1493178640625} > 0. \end{aligned}$$

This proves (5). $\square$

3 The seventeen rational localization faces

Let

$$\rho_c := \frac{1}{1+2\pi}.$$

Because $\pi > 3$, $\rho_c < 1/7$. The seventeen registry ratios are strictly ordered as follows:

$$\begin{aligned} &\frac{4}{25} < \frac{1}{5} < \frac{1}{4} < \frac{3}{10} < \frac{1}{3} < \frac{7}{20} < \frac{3}{8} < \frac{2}{5} < \frac{5}{12} < \frac{3}{7} < \frac{1}{2} \\ &\qquad < \frac{8}{15} < \frac{107}{200} < \frac{3}{5} < \frac{16}{25} < \frac{13}{20} < \frac{2}{3}. \end{aligned} \tag{6}$$

For a completely explicit distinctness check, the sixteen consecutive gaps in (6) are

$$\frac{1}{25}, \frac{1}{20}, \frac{1}{20}, \frac{1}{30}, \frac{1}{60}, \frac{1}{40}, \frac{1}{40}, \frac{1}{60}, \frac{1}{84}, \frac{1}{14}, \frac{1}{30}, \frac{1}{600}, \frac{13}{200}, \frac{1}{25}, \frac{1}{100}, \frac{1}{60}. \tag{7}$$

Moreover

$$\frac{4}{25} - \frac{1}{7} = \frac{3}{175} > 0, \qquad \frac{7}{8} - \frac{2}{3} = \frac{5}{24} > 0.$$

Thus all seventeen ratios lie in the high strip $\rho_c < \rho < 7/8$.

For a positive row below, we use

$$\Delta_{m;n,\ell}(r) > m(m+1) - \ell(\ell+1) - (n^2-1)\left(\frac{\pi_+}{1-r}\right)^2.$$

For a negative row, we use

$$-\Delta_{m;n,\ell}(r) > (n^2-1)\left(\frac{\pi_-}{1-r}\right)^2 - m(m+1) + \ell(\ell+1).$$

The resulting exact witnesses are the following. "Retain" means only that the product wall does not permit deletion of the indicated block; the independent Bessel wall may still remove it.

no.	r	$(m;n,\ell)$	exact sign witness	one-sided cap/event consequence
1	1/5	$(11;3,1)$	$\Delta > 320/49$	retain the third block 0:1 for $\rho \le 1/5$
2	3/10	$(8;2,3)$	$-\Delta > 58215/137641$	at k_8, $h \le 2$ for $\rho \ge 3/10$
3	1/3	$(8;2,2)$	$-\Delta > 27699/44944$	at k_8, $h \le 1$ for $\rho \ge 1/3$
4	3/8	$(9;2,3)$	$\Delta > 2622/1225$	retain the $h = 3$ column below this face
5	2/5	$(9;2,2)$	$\Delta > 248/147$	retain the $h = 2$ column below this face
6	5/12	$(9;2,1)$	$\Delta > 2200/2401$	retain the $h = 1$ column below this face
7	3/7	$(9;2,0)$	$-\Delta > 120843/179776$	no second mode at k_9 on the upper side
8	1/2	$(10;2,0)$	$-\Delta > 23677/2809$	no second mode at k_{10} on the upper side
9	107/200	$(11;2,0)$	$-\Delta > 13302732/2699449$	no second mode at k_{11} on the upper side
10	8/15	$(11;2,0)$	$-\Delta > 2175627/550564$	the clean stated cutoff: a second mode forces $\rho < 8/15$
11	3/5	$(8;2,0)$	$-\Delta > 5080707/44944$	select the no-second k_8 cap/payment side
12	16/25	$(9;2,0)$	$-\Delta > 1555635/11236$	select the no-second k_9 cap/payment side
13	13/20	$(10;2,0)$	$-\Delta > 18126190/137641$	select the no-second k_{10} cap/payment side
14	2/3	$(11;2,0)$	$-\Delta > 1510851/11236$	select the no-second k_{11} cap/payment side
15	4/25	$(10;3,0)$	$-\Delta > 260740/137641$	no third mode at k_{10} on the upper side
16	1/4	$(11;3,0)$	$-\Delta > 23484/2809$	a third mode at k_{11} forces $\rho < 1/4$
17	7/20	$(9;2,4)$	$-\Delta > 36230/474721$	the $H = 6, h = 4$ cap–74 cell can occur only below this face

Every entry is a displayed positive rational number. Together with the monotonicity (4), the table proves the asserted side on an interval, not merely at a sampled point.

4 Three algebraic splits and the five k_{11} implications

At $\rho = \rho_c$, $x = (\pi + 1/2)^2$, while at $\rho = 7/8$, $x = 64\pi^2$. Lemma 1 gives the exact chain

$$(\pi + \tfrac{1}{2})^2 < \left(\frac{51}{14}\right)^2 = \frac{2601}{196} < 16 < \frac{68}{3} < 34 < 64 \cdot 3^2 < 64\pi^2. \tag{8}$$

Consequently each of $x = 16, 68/3, 34$ is attained at a unique point in the high strip. The relevant channel arithmetic is

x	channel	exact value of Δ
16	$(m; n, \ell) = (11; 3, 1)$	$132 - 2 - 8 \cdot 16 = 2 > 0,$
68/3	$(10; 2, 6)$	$110 - 42 - 3(68/3) = 0,$
34	$(11; 2, 5)$	$132 - 30 - 3 \cdot 34 = 0.$

(9)

Thus the first split lies on the possible side. The latter two are exact channel walls, and their new modes are excluded by the strict convention.

The two localization consequences at k_{11} may also be read without the table:

$$x_{8/15} - 44 > \left(\frac{\pi_-}{1 - 8/15}\right)^2 - 44 = \frac{725209}{550564} > 0, \tag{10}$$

$$x_{1/4} - \frac{33}{2} > \left(\frac{\pi_-}{1 - 1/4}\right)^2 - \frac{33}{2} = \frac{5871}{5618} > 0. \tag{11}$$

Hence a second mode at k_{11} forces $\rho < 8/15$, and a third mode forces $\rho < 1/4$. Finally, the first-mode ball walls give

$$\left(\frac{69}{5}\right)^2 - 132 = \frac{1461}{25} = 44 + \frac{361}{25} = \frac{33}{2} + \frac{2097}{50}, \tag{12}$$

$$\left(\frac{64}{5}\right)^2 - 132 = \frac{796}{25} = \frac{33}{2} + \frac{767}{50}. \tag{13}$$

If $H = 10$ is present at k_{11}, then x is larger than both the second threshold 44 and the third threshold $33/2$; hence every second and third mode is excluded. If $H = 9$ is present, every third mode is excluded. Equations (10)–(13) are the five formerly separate k_{11} localization implications.

5 The three impossible cap cells

If the first mode H is counted at k_m, its strict ball wall $\beta_{H,1} < k_m$ forces $x > \beta_{H,1}^2 - m(m+1)$. If the second mode h is counted, its product wall forces $x < (m(m+1) - h(h+1))/3$. The three dashed cells are therefore empty by the following exact, incompatible bounds:

cell	required lower bound	required upper bound	gap
$k_8 : H = 6,\ h \geq 2$	$100 - 72 = 28$	$(72 - 6)/3 = 22$	6
$k_9 : H = 8,\ h \geq 0$	$(71/6)^2 - 90 = 1801/36$	$90/3 = 30$	721/36
$k_9 : H = 7,\ h \geq 3$	$(23/2)^2 - 90 = 169/4$	$(90 - 12)/3 = 26$	65/4.

(14)

No limiting equality can rescue a cell: both necessary modal inequalities are strict.

6 Checkpoint monotonicity and five global payments

Put $M = m(m+1)$ and $\mathcal{W}_m(\rho) := \mathcal{W}(\rho, k_m(\rho))$. Logarithmic differentiation shows that $\mathcal{W}_m'$ has the sign of

$$\pi^2(1 + \rho) - M\rho^2(1 - \rho)^2. \tag{15}$$

For $m \leq 11$,

$$\pi^2(1 + \rho) - M\rho^2(1 - \rho)^2 > 9 - \frac{132}{16} = \frac{3}{4} > 0.$$

Thus every $\mathcal{W}_m$, $m \leq 11$, is strictly increasing. This single argument replaces both executable derivative-bound predicates.

At ρ_c, one has $z = \pi + 1/2 > 193/53$, $\rho_c < 1/7$, and $1/\pi > 7/22$. Consequently

$$\mathcal{W}(\rho_c, K) > \frac{2}{9}\frac{7}{22}\left(1 - \frac{1}{7^3}\right) K^3 = \frac{38}{539}K^3. \tag{16}$$

For each row in the next table, the middle column proves $k_m(\rho_c) > t_m$, and the last column proves $\mathcal{W}_m(\rho_c) > C_m$.

m	cap C_m	t_m	$(193/53)^2 + M - t_m^2$	$(38/539)t_m^3 - C_m$
7	40	208/25	67049/1755625	5083656/8421875
8	53	923/100	1901439/28090000	656778873/269500000
9	73	254/25	61431/1755625	7911557/8421875
10	89	111/10	14211/280900	1999489/269500
11	121	241/20	65271/1123600	5076899/2156000

Every last-column entry is positive. Monotonicity therefore extends each global payment from ρ_c to the complete high strip.

7 Seven conditional first-block payments

We now give a general reduction which avoids any numerical enclosure of a radical. Suppose the presence of first mode H below the checkpoint k_m forces $K > t$. Set

$$a^2 := t^2 - m(m+1), \qquad \rho_0 := 1 - \frac{\pi}{a}. \tag{17}$$

All seven rows below have $a > \pi$: the smallest listed radicand is

$$\frac{89}{4}, \qquad \frac{89}{4} - \left(\frac{22}{7}\right)^2 = \frac{2425}{196} > 0.$$

If the mode is present for some $K \le k_m(\rho)$, then $k_m(\rho) > t$, hence $x_\rho > a^2$ and $\rho > \rho_0$. By the strict increase of $\mathcal{W}_m$,

$$\mathcal{W}_m(\rho) > \mathcal{W}_m(\rho_0) = \mathcal{W}(\rho_0, t).$$

Choose a rational $r > \rho_0$. Since $\mathcal{W}(\rho, t)$ decreases with ρ at fixed t, Lemma 1 yields

$$\mathcal{W}(\rho_0, t) > \mathcal{W}(r, t) > \frac{7}{99}(1 - r^3)t^3. \tag{18}$$

The condition $r > \rho_0$ follows, without taking a square root, from

$$L := \pi_-^2 - (1 - r)^2 a^2 > 0. \tag{19}$$

Here is the complete calculation. The column P is the reserve in (18) above the stated cap.

m	H	t	cap	a^2	r	L	$P = \frac{7}{99}(1 - r^3)t^3 - \text{cap}$
7	6	10	53	44	8/15	725209/2528100	18659/2673
8	7	23/2	80	241/4	3/5	64349/280900	213281/49500
9	8	71/6	81	1801/36	3/5	4713989/2528100	14506973/1336500
10	7	23/2	100	89/4	7/20	2105431/4494400	18539033/6336000

m	H	t	cap	a^2	r	L	P
10	8	71/6	97	1081/36	3/7	317585/4955076	2865431/261954
10	9	64/5	100	1346/25	3/5	8811001/7022500	25143284/1546875
11	9	64/5	125	796/25	1/2	536261/280900	58757/12375

All L's and P's are positive. Each row therefore proves both the localization of its conditional branch and its Weyl payment.

8 The connected cap–74 path

The seventeenth localization row is tied to the exceptional $k_9, H = 6, h = 4$ cap as follows. Angular propagation from the strict $(3, 2)$ Bessel wall gives

$$j_{9/2,2}^2 > \left(\frac{103}{10}\right)^2 + 20 - 12 = \frac{11409}{100} = 108 + \frac{609}{100}. \tag{20}$$

Accordingly the strict proxy uses $\beta_{4,2}^2 = 11409/100$; the strict inequality belongs to the actual Bessel zero $j_{9/2,2}$, not to the proxy constant. At k_9, the product wall for $(4, 2)$ is

$$3x < 90 - 20 = 70, \qquad \text{i.e.} \qquad x < \frac{70}{3}.$$

The ball wall (20), on the other hand, forces

$$x > \frac{11409}{100} - 90 = \frac{2409}{100}, \qquad \frac{2409}{100} - \frac{70}{3} = \frac{227}{300} > 0.$$

Thus the fully sharpened cap–74 cell is empty.

For completeness, even the deliberately conservative cell is paid. At $\rho = 7/20$,

$$x_{7/20} - \frac{70}{3} > \frac{400}{169}\left(\frac{333}{106}\right)^2 - \frac{70}{3} = \frac{36230}{1424163} > 0, \tag{21}$$

so a counted $(4, 2)$ mode forces $\rho < 7/20$. Also

$$108 - \left(\frac{207}{20}\right)^2 = \frac{351}{400} > 0, \tag{22}$$

so its ball wall forces $K > 207/20$. Therefore

$$\begin{aligned} \mathcal{W}(\rho, K) &> \frac{7}{99}\left(1 - \left(\frac{7}{20}\right)^3\right)\left(\frac{207}{20}\right)^3 \\ &= \frac{52823261673}{704000000} = 74 + \frac{727261673}{704000000} > 74. \end{aligned} \tag{23}$$

The cap itself is exactly $(6 + 1)^2 + (4 + 1)^2 = 49 + 25 = 74$. Equations (20)–(23) cover both the actual incompatibility and every row of the conservative cap–74 fallback.

9 Exact row map and scope audit

The following map uses the ordering of the two functions in the original exact registry. To keep the table readable, L denotes `verify_localizations`, and C denotes `verify_cap_tables_and_payments`. The map lets a referee match every replaced predicate to a displayed hand calculation.

registry family	row numbers	analytic replacement here
L	1–2	distinctness and connectedness: (6)–(7) and the seventeen-row table
L	3–19	all ratios in the high strip, by $\rho_c < 1/7 < 4/25$ and $2/3 < 7/8$
L	20–36	the seventeen exact sign witnesses in the long table
L	37–42	split location (8) and side/equality arithmetic (9)
L	43–47	the five k_{11} implications (10)–(13)
C	44–55	the twelve non-bookkeeping cap–74 predicates, (20)–(23); row $C56$, the block identity $49 + 25 = 74$, is allocated to Packet A
C	57–59	the three impossible cells, (14)
C	78–79	the general checkpoint monotonicity proof (15)
C	80–84	the five global-payment rows following (16)
C	85–98	the seven localization and seven conditional-payment rows in the final long table

The count reconciliation is

subfamily	connected mathematical objects	registry predicates
registry metadata	1	2
high-strip membership	17	17
localization signs	17	17
algebraic splits	3	6
special k_{11} implications	5	5
cap–74 path	1	12
impossible cells	3	3
checkpoint monotonicity	1	2
global payments	5	5
conditional payments	7	14
total	60	83.

In particular, the five headline Packet–C subfamilies contain exactly

$$17 + 3 + 3 + 5 + 7 = 35$$

connected objects (localization signs, algebraic splits, impossible cells, global payments, and conditional payments). Because each algebraic split has a location and a side predicate, and each conditional payment has a localization and a reserve predicate, these headline objects replace $17 + 6 + 3 + 5 + 14 = 45$ registry predicates. The remaining 38 predicates in the total are the explicitly listed auxiliary closure rows.

Thus this packet replaces all 47 predicates of L and 36 predicates of C. The elementary cap addition rows 1–43, row 56, and rows 60–77 of the latter family are outside the declared Packet C scope; they are instances of the general odd-sum/block-cap identity and are not claimed here. No row in the declared scope is unreconstructed.